



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	RESEARCH METHODOLOGY IN CHEMISTRY				
Type of Course	MAJOR				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. A strong grasp of foundational chemistry principles and terminology. 2. Understanding of basic research concepts.				
Course Summary	This course provides a comprehensive overview of research methodology in chemistry, covering the processes involved in conducting research, data analysis techniques, the role of computers in chemistry research, analytical techniques, scientific writing, and research ethics. Students will develop essential skills and knowledge to conduct research effectively and ethically in the field of chemistry.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the sequential processes involved in research, from topic formation to publication, encompassing hypothesis development, data collection, analysis, hypothesis revision, and effective communication of findings.	U	F	Instructor created exams / Quiz /Assignment
CO2	Develop proficiency in analyzing chemical data, including error classification, measurement accuracy, precision assessment, and statistical analysis application.	An	C	Class test /Assignment /Quiz
CO3	Acquire competency in utilizing computers for chemistry research, covering hardware, software, programming languages, operating	U	P	Assignment/ Class test

	systems, and specific applications like MS Office and scientific software.			
CO4	Apply various analytical techniques, such as chromatography, spectroscopy, electroanalysis, and thermal analysis, effectively in chemical research.	Ap	C	Assignments /Seminar presentation
CO5	Gain proficiency in scientific writing, including report structuring, language usage, and citation styles, while adhering to ethical standards like plagiarism avoidance and responsible data handling.	U	M	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Research Methodology		12	24
	1	Formation of the Topic	1	
	2	Hypothesis: Conceptual Definitions, Operational Definition	2	
	3	Gathering of Data, Analysis of Data, Revising of Hypothesis, Conclusion	3	
	4	Literature Survey: Journals, Books, and E-resources	3	
	5	Presentation and Publication of Research Output	3	
II	Data Analysis and Interpretation		12	24
	6	Errors in Chemical Analysis: Classification of Errors, Accuracy, Precision, and Reproducibility of Measurement	3	
	7	Methods of Analysis in Chemistry: Instrumental and Non-Instrumental	2	
	8	Presentation of Data: Mean, Standard Deviation	2	
	9	Comparison of Results: "t" Test, "f" Test, Chi-Square Test	2	
	10	Least Squares Analysis, Weighted Least Squares Analysis, Regression Coefficient, Rejection of Results	3	
III	Applications of Computers in Chemistry		12	24

	11	Types of Computers: Mainframe, Mini, Micro, Supercomputers, Personal Computers	2	
	12	Computer Hardware: CPU, Input and Output Devices, Memory, Peripheral Devices, Auxiliary Storage Devices	2	
	13	Computer Software: System Software, Application Software, Programming Languages: Machine Language, Assembly Language, High-Level Languages, Interpreter and Compiler	3	
	14	Operating Systems: Disk Operating System, Windows, macOS, Linux	2	
	15	Use of Internet in Research: Websites, Search Engines, e Journals, e-Libraries, INFLIBNET	1	
	16	Software Packages and Scientific Applications in Chemistry: Origin, Chems sketch, Chemdraw	2	
IV	Analytical Techniques for Chemical Research		12	26
	17	Chromatography: Thin layer chromatography, Column chromatography, Paper chromatography	2	
	18	Gas liquid chromatography, High pressure liquid chromatography (HPLC)	3	
	19	Spectroscopic methods: UV-Visible, IR,	1	
	20	NMR, Mass and ESR	2	
	21	Electroanalytic methods: Polarography, Coulometry, Cyclic voltammetry	2	
	22	Thermal analysis: thermogravimetry (TG)- differential thermal analysis (DTA) and differential scanning calorimetry (DSC)	2	
V	Scientific Writing and Ethics of Research		12	
	Open Ended	Significance of Report Writing, steps in Writing Report: Introduction, review of literature, scope, Materials and methods, Results and discussion, conclusions, Bibliography, Citation, Acknowledgements, Layout, Structure, and Language of typical reports, use of Illustrations, and tables, Overview of popular citation styles: APA, MLA, ASA, Chicago Manual of Style, Oral presentation: Planning, Preparation, Practice, Making presentation, Use of visual aids, Importance of effective communication. Environmental Impacts, Ethical Issues, Commercialization, Copyright, Intellectual Property Rights, Reproduction of Published Material, Plagiarism, Citation and Acknowledgement, Reproducibility, Accountability		

References:

1. Leedy, Paul D., Jeanne E. Ormrod, and Jeanne Ellis Ormrod. Practical Research: Planning and Design. Prentice Hall, 2004.
2. Graziano, Anthony M., and Michael L. Rau. Research Methods: A Process of Inquiry. Prentice Hall, 2006.
3. Smith, Robert V. Graduate Research: A Guide for Students in the Sciences. University of Washington Press, 1998.
4. Skoog, D. A., and M. West. Principles of Instrumental Analysis. Saunders Golden Sunburst Series.
5. Vogel, A. I. A Textbook of Quantitative Inorganic Analysis. ELBS Longman's Green and Co Ltd., London, 1962.
6. Jurs, Peter C. Computer Software Applications in Chemistry. 2nd ed., John Wiley & Sons, New York, 1996.
7. Madric, and Donevan. Understanding Computers. McGraw Hill.
8. Raman, KV. Computers in Chemistry. Tata McGraw Hill, 1993.
9. Wendlandt, WW. Thermal Methods of Analysis. Interscience, New York, 1964.
10. RA, DA. How to Write and Publish a Scientific Paper. Cambridge University Press, London, 1992.
11. Chandra, A., and T.P. Sexena. Style Manual. Metropolitan Book Company Ltd., New Delhi, 2000.
12. Bouchoux, D. E. Intellectual Property: The Law of Trademarks, Copyrights, Patents, and Trade Secrets

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	P O 6	P O 7
CO 1	2	1		2	2	2	2	2	1	1	2	2	2
CO 2	2	2		2	2	2	3	2	2	2	2	2	2
CO 3	2	2		2	2	2	2	2	2	3	2	1	2
CO 4	2	2	2	2	3	2	3	1	1	1	2	1	2
CO 5	2	1		2	2	2	3	2	3	2	2	2	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment/viva /seminar/Quiz	Project Evaluation	End Semester Examinations
CO 1	✓	✓ ✓		✓
CO 2	✓	✓✓		✓
CO 3	✓	✓		✓
CO 4		✓✓		✓
CO 5		✓✓		✓

ELECTIVE COURSES IN MAJOR